Ex : $\boxed{2.3}$

1) Optimisation de $f(x,y) = xy$
sous la contrainte $\underbrace{x-y+6}_{g(x,y)} = 0$

f et g sont de classe C^2 sur $\mathbb{R}^2$.

$$Dg(x,y) = \left(\frac{\partial g}{\partial x}(x,y) \quad \frac{\partial g}{\partial y}(x,y)\right) = \vec{\nabla} g(x,y)$$

ici la jacobienne des contraintes (une seule g)
est le gradient de g

rq : $g_1(x,y), g_2(x,y)$: 2 contraintes

$$Dg(x,y) = \begin{pmatrix} \frac{\partial g_1}{\partial x}(x,y) & \frac{\partial g_1}{\partial y}(x,y) \\ \frac{\partial g_2}{\partial x}(x,y) & \frac{\partial g_2}{\partial y}(x,y) \end{pmatrix} \begin{matrix} \to \vec{\nabla} g_1(x,y) \\ \to \vec{\nabla} g_2(x,y) \end{matrix}$$

On a $Dg(x,y) = \left(\frac{\partial}{\partial x}(x-y+6) \quad \frac{\partial}{\partial y}(x-y+6)\right)$

$$= \begin{pmatrix} 1 & -1 \end{pmatrix}$$

On définit le lagrangien L sur $\mathbb{R}^{2+1} = \mathbb{R}^3$

$\mathbb{R}^{n+p}$ avec $n = 2$, nb de variables de f
$p = 1$, nb de contraintes.

$$L(x,y,\lambda) = f(x,y) - \lambda g(x,y)$$
$$= xy - \lambda(x-y+6)$$

On a $\vec{\nabla} L(x,y,\lambda) = \left(\frac{\partial L}{\partial x}, \frac{\partial L}{\partial y}, \frac{\partial L}{\partial \lambda}\right) = \vec{0}$

$$\Leftrightarrow (y-\lambda, x+\lambda, -x+y-6) = (0,0,0)$$

$$\Leftrightarrow \begin{cases} y-\lambda = 0 \\ x+\lambda = 0 \\ -x+y-6 = 0 \end{cases} \Leftrightarrow \begin{cases} x = -\lambda \\ y = \lambda \\ \lambda+\lambda-6 = 0 \end{cases} \Leftrightarrow \begin{cases} x = -3 \\ y = 3 \\ \lambda = 3 \end{cases}$$

On a $X_0 = (-3,3)$ et $\lambda = 3 = \lambda_0$.

Vérification :
$Dg(X_0) = Dg(-3,3) = \begin{pmatrix} 1 & -1 \end{pmatrix}$
qui est bien de rang $p=1$
Ainsi les contraintes (ici 1) sont qualifiées.

On calcule la matrice Hessienne :

$$D^2 L(X_0,\lambda_0) = \begin{pmatrix} \frac{\partial^2 L}{\partial x^2}(X_0,\lambda_0) & \frac{\partial^2 L}{\partial x \partial y}(X_0,\lambda_0) \\ \frac{\partial^2 L}{\partial y \partial x}(X_0,\lambda_0) & \frac{\partial^2 L}{\partial y^2}(X_0,\lambda_0) \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\partial}{\partial x}(y-\lambda) & \frac{\partial}{\partial x}(x+\lambda) \\ \frac{\partial}{\partial y}(y-\lambda) & \frac{\partial}{\partial y}(x+\lambda) \end{pmatrix} \Bigg|_{\substack{x=-3 \\ y=3 \\ \lambda=3}}$$

$$= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = H$$

↳ valeurs propres : $\det(H - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - 1 = (\lambda-1)(\lambda+1)$

$sp(H) = \{+1, -1\}$ on ne peut pas conclure.

Pour $X_0 = (-3,3)$ et $\lambda_0 = \lambda = 3$ → matrice par bloc
On construit la Hessienne bordée
carrée d'ordre $n+p = 2+1 = 3$.

$$H(X_0) = \begin{pmatrix} \underbrace{[0]}_{(p,p)} & \underbrace{Dg(X_0)}_{(p,n)} \\ \underbrace{(Dg(X_0))^T}_{(n,p)} & \underbrace{D^2L(X_0,\lambda_0)}_{(n,n)} \end{pmatrix}$$

On a $H(-3,3) = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}$

$$\det(H(-3,3)) = \begin{vmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & \boxed{2} & 0 \\ \boxed{1} & 0 & \boxed{1} \\ \boxed{-1} & 1 & \boxed{0} \end{vmatrix} \; L_1+L_2+L_3$$

$$= -2(0+1) = -2$$

$\left.\begin{matrix} n=2 \\ p=1 \end{matrix}\right\}$, $\det(H(-3,3))$ est du signe de $(-1)^p = (-1)^1 = -1$
et non de $(-1)^n = (-1)^2 = 1$

Conl f admet un minimum local en $X_0 = (-3,3)$
sous la contrainte $g(x,y) = 0 \Leftrightarrow x-y+6 = 0$